

Wisdom Investing

Intelligent Portfolio Management System

Foundational Architecture Decisions

Step 3 Response -- Technical Architecture Brief

Prepared for Senior Leadership | Investment Management

June 2026 | Confidential

At a Glance

Portfolio universe:	4 high-conviction NSE-listed equities (AMBER, DBL, WELSPUNLIV, ZEEL)
System type:	Quantitative, rule-based; LLM used for development tooling only
Decision frequency:	Daily signals; weekly stop-loss recalibration; monthly price ceilings
Backtest CAGR:	16.72% (optimised) vs 6.5% baseline over 10-year window 2015-2025
Sharpe / Max Drawdown:	Sharpe 0.41 Max Drawdown -36.6% (pre-optimisation)
Deployment:	Wisdom Trader -- password-gated Streamlit dashboard

This document describes the architecture of a working prototype built as part of the Wisdom Investing AI Engineer Skills Assessment. All data is proprietary and used solely for prototype development.

1. Executive Overview

The Wisdom Trader system is a quantitative, rules-based intelligent portfolio manager built for a concentrated portfolio of four NSE-listed equities held by Wisdom Investing (AMBER, DBL, WELSPUNLIV, ZEEL). The system ingests 10 years of daily OHLCV data, constructs a multi-timeframe feature set, applies a four-condition buy-signal filter, and executes a daily backtesting loop that simulates real investment decisions under realistic constraints: beginning-of-day portfolio freeze, stop-loss discipline, allocation caps, and risk-calibrated position sizing.

Hyperparameter optimisation using Bayesian TPE search across 200 trials lifted the simulated CAGR from a 6.5% baseline to 16.72%, a result that compares favourably to long-term Nifty 50 total returns of approximately 14-15% p.a. over the same window. The system surfaces daily buy/sell recommendations and backtest analytics through the Wisdom Trader dashboard -- a password-gated web application designed for portfolio managers to interrogate signals and experiment with strategy parameters.

This note addresses the five architecture pillars prescribed in Step 3: data pipelines, data security and privacy, LLM usage, AI agents, and cloud deployment. It is written for senior investment management leadership and intentionally avoids implementation minutiae in favour of design rationale and risk considerations.

2. Data Pipeline Architecture

2.1 Data Sources and Ingestion

The prototype operates on three raw data streams, each sourced manually for the four-stock portfolio:

Data Stream	Source	Granularity	Refresh Cadence
OHLCV price data	NSE / BSE / Yahoo Finance	Daily bars	Weekly (manual)
Trade history	WISDOM tradebook (Excel)	Per transaction	One-time load
Analyst research	Company IR pages, Screener	Periodic reports	On publication

Raw OHLCV files are stored as per-ticker CSVs in data/ohlc/. The ingestion function concatenates all files, deduplicates on (date, symbol), and sorts by (symbol, date) to produce a clean daily price frame. The tradebook is an encrypted Excel file accessed via Azure Key Vault-managed credentials, ensuring the proprietary transaction history never traverses the network in cleartext.

2.2 Multi-Timeframe Feature Engineering

The engineering pipeline computes a 65-column daily feature set from the raw OHLCV frame through four sequential stages:

- Daily features -- ATR(14), EMA(21), SMA(30), RSI(7), Connors RSI (CRSI), Supertrend, daily net-change normalised by ATR. These capture momentum, volatility, and short-term trend direction.
- Weekly aggregation (W-FRI) -- OHLCV bars resampled to week-ending-Friday. Weekly ATR(14), SMA(30), weighted-DMA(30), and Supertrend are computed. Stop-loss level (sl) is constructed as the minimum of three floors: $(\min(\text{SMA30}, \text{WDMA30}) - \text{atr_mult} \times \text{ATR14}), (1 - \text{st_buffer}) \times \text{Supertrend_current}$, and $(1 - \text{st_buffer}) \times \text{Supertrend_prior}$. This multi-floor design ensures the stop is always conservative, never above the trend-floor.
- Monthly aggregation (ME) -- Monthly OHLCV. Swing peak detection (scipy argrelextrema) identifies prior highs. Max allowable buy price is set at $\text{overshoot_coeff} \times \text{prior_peak}$, preventing entries into speculative momentum above a configurable ceiling.
- Cross-timeframe mapping -- Weekly stop-loss values are forward-filled onto daily dates. Monthly price ceilings are similarly mapped. This preserves weekly/monthly discipline in a daily signal framework without look-ahead bias.

2.3 Signal Computation

The four-condition buy flag requires all of the following to be simultaneously true:

Condition	Indicator	Rationale
Trend filter	Close > Weekly Supertrend	Only buy in a confirmed weekly uptrend
Momentum gate	CRSI < crsi_threshold (~44-50)	Avoid overbought entries; buy on pullbacks
Risk gate	Stop-loss risk < risk_threshold	Reward/risk ratio is acceptable
Price ceiling	Close < monthly peak x coeff	Avoids chasing parabolic moves

The sell signal is unambiguous: close price at or below the weekly stop-loss. There is no discretionary override -- the stop-loss is structural and always enforced, consistent with WISDOM's stated discipline of cutting losers early.

2.4 Output Artefacts

- features_d.csv / .parquet -- Full daily feature set, the primary input to the engine
- trade_log -- Every BUY and SELL event with execution price, units, portfolio state
- daily_signals -- Per-ticker, per-day snapshot: in/out of portfolio, buy/sell flags, estimated position size
- portfolio_daily -- End-of-day mark-to-market value, cash, invested value, cumulative return
- trade_summary -- Closed-trade P&L: entry/exit dates, average buy price, realised gain

3. Trading Engine Architecture

3.1 Core Design Principles

The engine follows a strict daily simulation loop over 10 years of data. Three design decisions differentiate it from naive backtesting frameworks and are worth highlighting for an investment management audience:

Decision	Implementation	Why It Matters
BOD portfolio freeze	Portfolio value fixed at market open; all sizing d	Eliminates intraday circular dependency; mirrors actual pre-market work
Sell before buy	Stop-loss exits processed before any new entr	Frees capital for reinvestment; reflects operational reality
Mark-to-market returns	CAGR and drawdown include open positions v	Total return, not just realised P&L; consistent with fund accounting

3.2 Risk-Based Position Sizing

Every buy is sized to risk exactly MAX_RISK_PER_TRADE (e.g., 1-4%) of portfolio value, defined as the capital lost if the stop-loss is hit immediately:

$$\text{Position Size} = (\text{MAX_RISK_PER_TRADE} \times \text{Portfolio Value}) / \text{Stop-Loss \%}$$

$$\text{Units} = \text{floor}(\text{Position Size} / \text{Current Close})$$

This approach -- sometimes called fixed-fractional or Kelly-adjacent sizing -- ensures that no single losing trade can inflict disproportionate damage on the portfolio, regardless of the volatility of the individual security. A secondary cap (MAX_SECURITY_ALLOC, e.g., 33-90%) prevents over-concentration in any single name, enforcing portfolio-level diversification.

3.3 Add-On Buy Logic

For securities already held, a second buy is permitted only if the current price is above either the last buy price or the 21-day EMA. This momentum gate prevents averaging down into falling positions -- a common behavioural trap identified in WISDOM's historical trade analysis -- while allowing pyramid-style position building into confirmed strength.

3.4 Performance Baseline

Metric	Baseline (default params)	Optimised (Trial #348)
CAGR	6.53%	16.72%
Cumulative Return	105.8%	~350%+

Sharpe Ratio	0.408	Higher
Max Drawdown	-36.6%	Reduced
Total Trades	15	More
Win Rate	33.3%	Higher

4. Hyperparameter Optimisation Architecture

4.1 Framework: Optuna with TPE Sampler

Strategy parameters that are difficult to set by intuition alone -- stop-loss sensitivity, risk tolerance, momentum thresholds -- are optimised using Optuna, a Bayesian optimisation framework. The Tree-structured Parzen Estimator (TPE) sampler models the joint distribution of parameters that produced high CAGR outcomes and samples new trials accordingly, converging far faster than random or grid search.

A key architectural choice: all heavy indicator computations (Supertrend, ATR, RSI, CRSI) are run once before the search begins and cached in memory. Each trial only recomputes the lightweight arithmetic that depends on the trial's parameters (stop-loss floors and buy-price ceilings), reducing per-trial latency to 3-4 seconds and making 200-trial searches practical on a laptop.

4.2 Parameter Search Space

Parameter	Range	Controls	Optimal Value
max_security_alloc	[0.20, 0.90]	Per-name concentration cap	0.90
max_risk_per_trade	[0.005, 0.05]	Capital at risk per entry	0.038
atr_mult	[0.20, 3.00]	ATR contribution to weekly SL floor	2.37
st_buffer	[0.005, 0.25]	% below Supertrend for SL	0.118
overshoot_coeff	[0.90, 1.30]	Max entry as multiple of peak	1.22
crsi_threshold	[1, 75]	CRSI upper bound for entries	44.5
risk_threshold	[0.05, 0.60]	Max stop-loss % to permit entry	0.287

4.3 Interpretation of Optimal Parameters

The optimal solution for this portfolio and time window reveals several investment insights:

- High concentration tolerance (alloc 0.90): for a 4-stock portfolio managed with conviction, the model confirms that concentration -- not diversification -- drives returns. This aligns with WISDOM's stated high-conviction philosophy.
- Larger position risk (3.8% per trade vs 1% default): the optimiser found that the quality of signals in this universe warranted larger bets when conditions aligned, rather than cautious fractional sizing.
- Tighter momentum filter (CRSI < 44.5): entries are restricted to more oversold conditions, improving entry timing and reducing the frequency of buying into short-lived rallies.
- Wider ATR stop-floor (atr_mult 2.37): stops are set further below moving averages, tolerating more short-term volatility before exiting -- consistent with WISDOM's 10-20 year holding horizon.

5. Dashboard and Deployment Architecture

5.1 Wisdom Trader Dashboard

The Wisdom Trader is a three-tab Streamlit web application providing portfolio managers with an interactive interface to the engine. It is pre-loaded with the latest features_d.csv on startup and requires no data upload step for day-to-day use.

Tab	Function	Key Controls
Daily Signals	Calendar date picker -- shows buy recommendations and stop-loss alert	Date picker; portfolio-aware sizing after backtest run
Backtest	Full simulation with portfolio curve, drawdown chart, closed-trade P&L, etc.	Run Backtest button; ticker selector for chart
Parameters	All 7 optimised parameters exposed as sliders; Apply & Run triggers full re-optimisation	Load Optuna Best; Apply & Run

5.2 Deployment Options

Three deployment tiers are available, in ascending order of infrastructure commitment:

Tier	Method	Suitable For	Infrastructure
1 -- Local tunnel	ngrok exposes localhost:8501 via public HTTPS	Demo sharing, short-term POC	None -- runs on analyst laptop
2 -- Managed cloud	Streamlit Community Cloud (free tier, GitHub-backed)	Persistent POC, team access	No servers; GitHub repo only
3 -- Private cloud	AWS EC2 t3.small with nohup; port 8501 behind	Production POC, full pipeline	One EC2 instance (~\$15/month)

For the current prototype, Tier 1 (ngrok) or Tier 2 (Streamlit Community Cloud) are recommended. Tier 3 is required only if the full engineering pipeline rerun capability (ATR multiplier, Supertrend buffer, and other weekly/monthly parameters) needs to be available from the dashboard, as that requires the raw OHLCV data to reside on the server.

5.3 Authentication

A password gate (SHA-256 hashed, environment-variable configured) guards the dashboard. For a production deployment, this would be replaced with the firm's existing SSO provider (e.g., Entra ID / Azure AD) using OAuth 2.0, which would also enable per-user audit logs of strategy parameter changes -- critical for compliance in a regulated environment.

6. Data Security and Privacy

6.1 Data at Rest

- Raw OHLCV and tradebook data are stored locally on the analyst's machine or the EC2 instance -- never in a public S3 bucket or shared cloud storage in the current prototype.
- The tradebook (proprietary transaction history) is encrypted at rest as a password-protected Excel file. The password is retrieved at runtime from Azure Key Vault, never hardcoded.
- Feature artefacts (features_d.csv, trade_log.parquet) are derived data and contain no client-identifiable information beyond ticker symbols.

6.2 Data in Transit

- Azure Key Vault communication uses TLS 1.2+ enforced by the Azure SDK.
- Dashboard access is over HTTPS when deployed via Streamlit Community Cloud or behind an EC2 load balancer with an ACM certificate. ~~The ngrok tunnel also terminates TLS.~~
- No raw price data or trade data is transmitted to any external API in normal operation. The LLM (Claude) was used exclusively as a development coding assistant; it received code and feature logic, not proprietary transaction records.

6.3 LLM Data Handling

This is a critical boundary: the trading system itself contains no LLM calls. The LLM (Anthropic Claude via Claude Code) was used as an AI pair-programmer to build, debug, and optimise the codebase. No proprietary OHLCV data, trade history, or client-sensitive information was shared with the LLM. Code, feature engineering logic, and strategy design were shared -- these are methodology artefacts, not data artefacts.

7. LLM Usage

7.1 Where the LLM Sits

In the current prototype, the LLM is entirely outside the production decision loop. Every buy signal, stop-loss, position size, and portfolio allocation is computed deterministically from mathematical rules. This was a deliberate architectural choice: for a regulated investment context, explainability and auditability are non-negotiable, and a rules-based engine satisfies both requirements without requiring hallucination guardrails or output validation.

7.2 Production Extension: Where LLM Adds Value

The architecture is designed to accept LLM augmentation in two well-bounded roles without compromising the deterministic signal engine:

LLM Role	Mechanism	Guardrail
Qualitative signal filter	RAG over analyst reports, earnings call transcripts, and annual reports. LLM a LLM output is advisory only; cannot override	
Portfolio commentary	Natural language summary of daily signals and portfolio state for the PM. Surf All factual claims grounded in retrieved position	

The LLM is never given discretionary power over position sizing, entry, or exit. Human-in-the-loop oversight is preserved: the PM sees the signal, the LLM commentary, and the source documents before making any action.

7.3 Hallucination Risk Management

- Ground all LLM responses in retrieved context (RAG); no unconstrained generation on financial metrics.
- Require source citations for any numerical claim (EPS, revenue, price target).
- Use temperature = 0 for all production inference to maximise determinism.
- Log all LLM inputs and outputs for compliance audit trail.
- Rate-limit LLM calls to one per signal event; prohibit chained agentic loops in production.

8. AI Agents

The prototype does not use autonomous AI agents in the production signal path. The Optuna optimisation framework is the closest analogue -- it autonomously explores the parameter space over 200 trials and selects the best configuration -- but it operates offline, not in real-time, and its outputs are always reviewed by the analyst before being applied.

For a production system, agentic capabilities would be introduced cautiously and constrained to two patterns:

- Data refresh agent -- scheduled to fetch new OHLCV data, recompute features, and regenerate the features_d artefact nightly. Operates on data, not capital.
- Research ingestion agent -- monitors company IR pages and Screener.in for new filings, downloads and chunks them into the RAG vector store. Triggered by calendar, not by market events.

Neither agent touches the trading engine or has access to order execution. The boundary between data infrastructure (agent-permissible) and capital allocation (human-only) is a hard architectural constraint, not a soft guideline.

9. Cloud Architecture

9.1 Recommended Anchor Services

For a production deployment on AWS, two services anchor the architecture:

Service	Role	Rationale
AWS EC2 (t3.small / t3.medium)	Hosts the Streamlit dashboard, stores OHLCV data and feature artefact	Simple, auditable, no serverless cold-start latency; cost-effective
AWS S3	Versioned storage for features_d artefacts and backtest outputs; option: On-premise dataset	Durability, versioning (recover any production dataset)

9.2 Supporting Services (Thin Layer)

- Azure Key Vault -- already in use for tradebook credentials; retain for consistency.
- AWS Secrets Manager -- alternative for dashboard password and any API keys if moving fully to AWS.
- AWS CloudWatch -- monitoring and alerting for EC2 instance health and nightly job failures.
- ACM + ALB -- for HTTPS termination and a custom domain (e.g., trader.wisdom.in) in a production deployment.

9.3 Scalability Path

The current prototype is intentionally simple: one EC2 instance, one Python process, one portfolio. The architecture scales in clearly defined steps:

Horizon	Change	What It Enables
Near-term	Expand OHLCV universe to 20-50 stocks	Broader opportunity set; feature pipeline scales linearly with data
Medium-term	Move backtest and optimisation to AWS Batch or Lambda	Parallel Optuna trials; reduce optimisation wall-clock from 1h to 10m
Long-term	Add RAG layer (Bedrock Knowledge Base or Pinecone) for research	LLM-augmented qualitative filters alongside quantitative signals

10. Summary and Next Steps

The Wisdom Trader prototype demonstrates that a disciplined, quantitative portfolio management system -- built on transparent rules, rigorous backtesting, and Bayesian optimisation -- can be assembled rapidly with modern AI-assisted development tooling and deployed as a usable decision-support interface for portfolio managers.

The three highest-priority architectural investments for moving from prototype to production are:

- Automated data refresh -- nightly OHLCV fetch, feature recomputation, and artefact versioning to S3. Eliminates the current manual upload step.
- SSO authentication -- replace the password gate with Azure AD / Entra ID to enforce access control, enable per-user audit logs, and satisfy compliance requirements.
- RAG research layer -- ingest analyst reports and company filings into a vector store; surface LLM-generated qualitative commentary alongside quantitative signals in the Daily Signals tab. This bridges the quantitative and fundamental analysis disciplines that characterise WISDOM's investment philosophy.

All code, backtest outputs, optimisation results, and this document are available in the GitHub repository accompanying this submission.